LESSON 02

Clean and transform with PySpark

Eight fictional orders. Choose columns, reject invalid rows, label missing text, derive a decimal total, and save a new managed Delta table with mode("error").

Cleaning rules

- select builds a new DataFrame without internal_note. raw_orders still has that column and still has 8 rows.
- Do not fill a missing quantity or price with zero. That would invent a sale.
- Keep a row only when order_id, quantity, and unit_price are present, quantity > 0, and unit_price >= 0. A zero price is allowed. A negative quantity is not a return in this lesson.
- Labels use trim. Null or blank becomes Unknown. Region is then uppercased. The helper returns a Spark column expression, not a Python UDF.
- order_total is quantity * unit_price, cast to decimal(12,2). orderBy affects the display, not the physical order of the saved table.

```
valid = (
    F.col("order_id").isNotNull()
    & F.col("quantity").isNotNull()
    & F.col("unit_price").isNotNull()
    & (F.col("quantity") > 0)
    & (F.col("unit_price") >= 0)
)
accepted = selected.filter(valid)
rejected = selected.filter(~valid)
```

```
def label_or_unknown(column_name):
    value = F.trim(F.col(column_name))
    return F.when(
        value.isNull() | (value == ""), F.lit("Unknown")
    ).otherwise(value)

clean_orders = (
    accepted
    .withColumn("customer", label_or_unknown("customer"))
    .withColumn("region", F.upper(label_or_unknown("region")))
    .withColumn(
        "order_total",
        (F.col("quantity") * F.col("unit_price")).cast("decimal(12,2)"),
    )
)
```

What is wrong in the eight raw rows

id	raw issue
1001	customer is ' Ana ' with surrounding spaces
1002	quantity is null — reject, do not invent 0
1003	customer is null — later labeled Unknown
1004	quantity is -1 — reject
1005	unit_price is null — reject, do not invent a free sale
1006	valid; internal_note is null and is not selected
1007	valid west order
1008	region is blank — later labeled UNKNOWN

Safe Delta write

- Schema name is bt_clean_orders_ plus 12 hex characters from uuid4, in the current catalog. A full rerun creates another schema. It does not replace the previous one.
- CREATE SCHEMA, then one managed table clean_orders. No LOCATION, no /mnt, no DROP, no spark.conf.
- Write with format("delta").mode("error"). That fails if the table already exists. Do not change it to overwrite.
- Keep the spelling error. On Free Edition Spark Connect, the alias errorifexists was rejected during serialization, before a table was created.
- If the write cell has already succeeded, skip it and run the SQL check. The notebook does not delete the demo schema for you.

```
catalog = spark.sql("SELECT current_catalog()").first()[0]
demo_schema = "bt_clean_orders_" + uuid4().hex[:12]
table_fqn = f"`{catalog}`.`{demo_schema}`.`clean_orders`"
# CREATE SCHEMA {quoted schema} COMMENT '...fictional orders demo'
clean_orders.write.format("delta").mode("error").saveAsTable(table_fqn)
```

Verify the stored table

Read the fully qualified table. In-memory checks and the SQL summary both expect 5 rows and total_sales 66.00. Persisted order_id values are 1001, 1003, 1006, 1007, 1008.

```
summary = spark.sql(f"""
SELECT COUNT(*) AS row_count,
       SUM(order_total) AS total_sales
FROM {table_fqn}
""")
# row_count == 5
# total_sales == Decimal("66.00")
```

id	customer	region	qty	price	total
1001	Ana	NORTH	2	10.00	20.00
1003	Unknown	EAST	3	7.50	22.50
1006	Fay	NORTH	1	12.00	12.00
1007	Gus	WEST	2	4.25	8.50
1008	Hana	UNKNOWN	1	3.00	3.00

Rejected rows stay out of the table. Inspect them; do not zero-fill them.

id	customer	why it is rejected
1002	Bo	quantity is null
1004	Dee	quantity is -1
1005	Eli	unit_price is null

Free Edition run: 8 raw, 5 clean, 3 rejected, total 66.00. All 8 code cells passed. Ana is trimmed from ' Ana '. The blank region on 1008 becomes UNKNOWN.

Practice, without writing: NORTH rows with order_total >= 15, columns order_id, customer, order_total. Expected row: 1001, Ana, 20.00.